

# DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

## Supplementary Summer Examination – 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: III

Subject Code & Name: BTCVC304      Hydraulics-I

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hr.

### Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)      Marks

|             |                                                                                                                                                                          |     |           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                                                   |     | <b>12</b> |
| A)          | Define following fluid properties with short explanation:- specific weight, specific volume, specific gravity, viscosity, surface tension, capillarity.                  | CO2 | 6         |
| B)          | State and prove Pascal's law with neat diagram.                                                                                                                          | CO2 | 6         |
| C)          | Define buoyancy and metacentric height. Derive an expression for the metacentric height of a floating body.                                                              | CO2 | 6         |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                   |     | <b>12</b> |
| A)          | The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as<br>$u = 2x + \frac{y^3}{3} - yx^2 ; v = -2y - \frac{x^3}{3} + xy^2$ | CO1 | 6         |
|             | (a) Obtain an expression for stream function $\Psi$                                                                                                                      |     |           |
|             | (b) Obtain an expression for velocity potential $\Phi$                                                                                                                   |     |           |
| B)          | Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation.                                                                           | CO1 | 6         |
| C)          | Discuss briefly the different methods of drawing the flow net.                                                                                                           | CO1 | 6         |
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                   |     | <b>12</b> |
| A)          | Derive an expression for mean velocity for laminar flow<br>(i) through a pipe,<br>(ii) between two parallel plates-both plates at rest.                                  | CO4 | 6         |
| B)          | Write a short note on Nikuradse's experiment.                                                                                                                            | CO4 | 6         |
| C)          | Explain the boundary layer theory.                                                                                                                                       | CO4 | 6         |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                   |     | <b>12</b> |
| A)          | Explain methods of dimensional analysis.                                                                                                                                 | CO4 | 6         |

|                                             |                                                                                                  |     |           |
|---------------------------------------------|--------------------------------------------------------------------------------------------------|-----|-----------|
| B)                                          | Explain different types of similarities that must exist between a prototype and its model.       | CO4 | 6         |
| C)                                          | Explain the terms: distorted models and undistorted models. What is the use of distorted models? | CO4 | 6         |
| <b>Q. 5 Solve Any Two of the following.</b> |                                                                                                  |     | <b>12</b> |
| A)                                          | What is syphon? Explain on what principle it works?                                              | CO3 | 6         |
| B)                                          | Discuss the concept of Water Hammer and Surge Tank.                                              | CO3 | 6         |
| C)                                          | Explain the Nomogram and Moody's chart.                                                          | CO3 | 6         |

\*\*\* End \*\*\*